# How to isolate x and t in v = x / t

## I want x

1. Write the formula. $$v = \frac{x}{t}$$

2. **t** is **dividing** **x**: multiply **both sides** by t. $$v \cdot t = \frac{x \cdot \cancel{t}}{\cancel{t}}$$

✓ Result: $$\boxed{x = v \cdot t}$$

## I want t

1. Start from x = v · t (left card). $$x = v \cdot t$$

2. **v** is **multiplying** **t**: divide **both sides** by v. $$\frac{x}{v} = \frac{\cancel{v} \cdot t}{\cancel{v}}$$

✓ Result: $$\boxed{t = \frac{x}{v}}$$

**Shortcut:** what is **dividing** goes to the other side **multiplying**, and what is **multiplying** goes to the other side **dividing**.